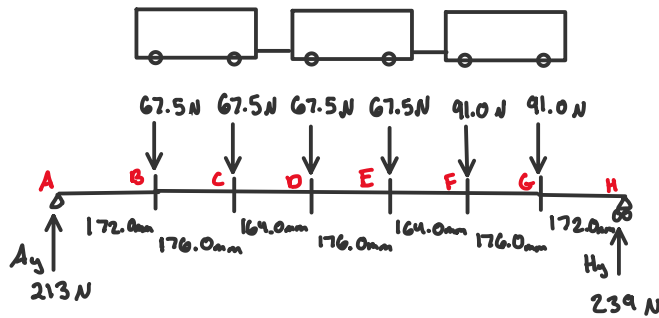


CIV102 Project Team 810 Deliverable 1

Monday, November 10, 2025 3:25 PM

1. & 2.



Total weight of train = 452 N

Freight Car = 135.0 N

Evenly distributed between two axes therefore $F = 135 / 2 = 67.5$

Locomotive Car = 182.0 N

Evenly distributed between two axes therefore $F = 182 / 2 = 91.0$ Effective length of train (from point forces) = $176 + 164 + 176 + 164 + 176 = 856$ mmLength of sides = $1200 - 856 = 344$ mm2 equal sides therefore each side is $344 / 2 = 172.0$ mm $\Sigma M = 0$

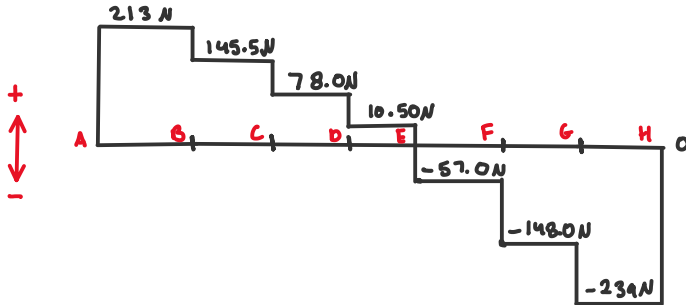
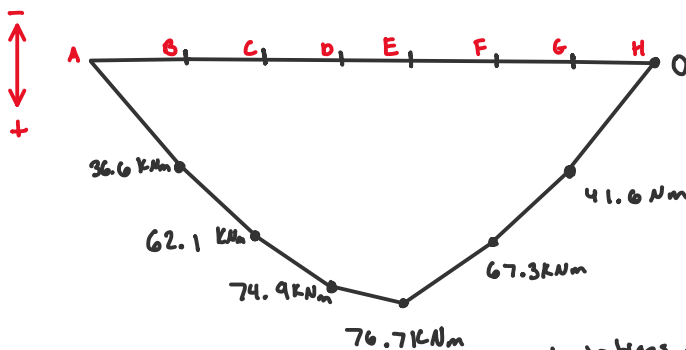
$$0 = 172(-67.5) + 348(-67.5) + 512(-67.5) + 688(-67.5) + 852(-91) + 1028(-91) + 1200H_y$$

$$239.3166 = H_y = 239 \text{ N}$$

 $\Sigma F_y = 0$

$$0 = A_y - 67.5 - 67.5 - 67.5 - 67.5 - 91 - 91 + 239.3166$$

$$212.6834 = A_y = 213 \text{ N}$$

3. SFD: $V(x)$ BMD: $M(x)$ SFD $\rightarrow$ BMD

Calculations done with unrounded answers:

Area AB = $213 \times 0.172 = 36.6 \text{ kNm}$ Area AC = $(145.5 \times 0.176) + 36.6 = 62.1 \text{ kNm}$ Area AD = $(78 \times 0.164) + 62.1 = 74.9 \text{ kNm}$ Area AE = $(10.5 \times 0.176) + 74.9 = 76.7 \text{ kNm}$ Area AF = $(-57 \times 0.164) + 76.7 = 67.3 \text{ kNm}$ Area AG = $(-148 \times 0.176) + 67.3 = 41.6 \text{ kNm}$ Area AH = $(-239 \times -0.172) + 41.6 = 0 \text{ kNm}$

calculations done with unrounded values:

$$\bar{y} = \frac{\sum y_i A_i}{\text{total area}}$$

distance from base of cross-section to the centroidal axis of each numbered section

 $A = b \cdot h$

① $100 \times 1.27 = 127.0 \text{ mm}^2$

② $62.7 \times 1.27 = 79.6 \text{ mm}^2$

③ $75 \times 1.27 = 95.25 \text{ mm}^2$

④ $80 \times 1.27 = 101.6 \text{ mm}^2$

⑤ $100 \times 1.27 = 127.0 \text{ mm}^2$

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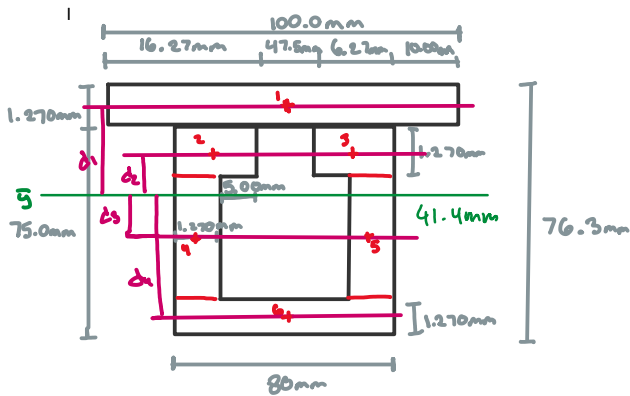
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about its own centroid
 $I_{total} = I_1 + A_1 d_1^2 + \dots$

$$I_x = \frac{bh^3}{12}$$

$$① \frac{100 \cdot 1.27^3}{12} = 17.07 \text{ mm}^4$$

$$② \frac{(1.27+5)(1.27)^3}{12} = 1.070 \text{ mm}^4$$

$$③ \frac{(1.27+5)(1.27)^3}{12} = 1.070 \text{ mm}^4$$

$$④ \frac{(1.27)(75-2(1.27))^3}{12} = 40560.87 \text{ mm}^4$$

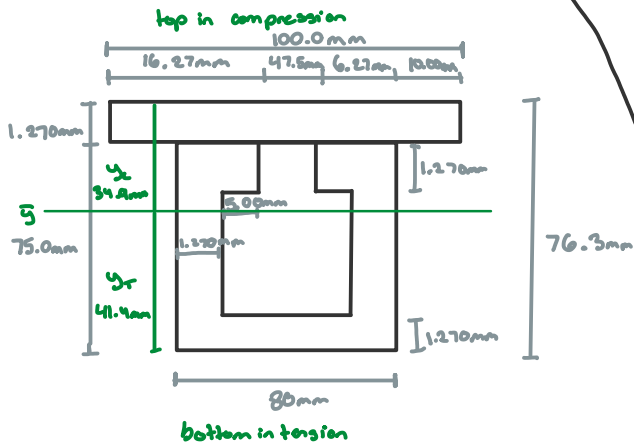
$$⑤ \frac{(1.27)(75-2(1.27))^3}{12} = 40560.87 \text{ mm}^4$$

$$⑥ \frac{(80)(1.27)^3}{12} = 13.66 \text{ mm}^4$$

$$\text{Total: } 80560.87 \text{ mm}^4$$

$$= 80.6 \times 10^3 \text{ mm}^4$$

5. FOS – Flexural Tension/Compression



$$d_i^2 = (\bar{y}_i - \bar{y})^2$$

$$① (75.6 - 41.4)^2 = 1172 \text{ mm}^2$$

$$② (74.4 - 41.4)^2 = 1087 \text{ mm}^2$$

$$③ (37.5 - 41.4)^2 = 15.21 \text{ mm}^2$$

$$④ (0.635 - 41.4)^2 = 1662 \text{ mm}^2$$

$$A_i d_i^2$$

I-parallel axis:

$$① 127 \times 1172.03 = 148847.81 = 148.8 \times 10^3 \text{ mm}^4$$

$$② 863.214 = 86.3 \times 10^3 \text{ mm}^4$$

$$③ 1399.68 = 1400 \text{ mm}^4$$

$$④ 168837.4 = 168.8 \times 10^3 \text{ mm}^4$$

$$\text{Total: } 33791.7 \text{ mm}^4$$

$$= 33.8 \times 10^3 \text{ mm}^4$$

$$\text{Total } I = 80560.87 + 33791.7 = 418352.57$$

$$\therefore I_{total} = 0.418 \times 10^6 \text{ mm}^4$$

$$\text{Safety factor} = \frac{\text{Average strength}}{\text{Average applied load}}$$

$$\text{Tension SF} = \frac{\sigma_t'}{\sigma_{tension}} = \frac{30}{7.6} = 3.9473$$

$$\text{Compression SF} = \frac{\sigma_c'}{\sigma_{compression}} = \frac{6}{6.39} = 0.9389...$$

$$\sigma_{tension} = \frac{M_y}{I}$$

$$\sigma_{tension} = \frac{(76.7 \times 10^3)(41.4)}{0.418 \times 10^6}$$

$$\sigma_{tension} = 7.5866... = 7.60 \text{ MPa}$$

$$\sigma_{compression} = \frac{M_y}{I}$$

$$\sigma_{compression} = \frac{(76.7 \times 10^3)(34.9)}{0.418 \times 10^6}$$

$$\sigma_{compression} = 6.389... = 6.39 \text{ MPa}$$

$$\text{FOS tension} = 3.95$$

$$\text{FOS compression} = 0.939$$

